```python
from pyspark.sql import SparkSession, DataFrame
from pyspark.sql.types import StructType, StructField, DoubleType,
StringType
import pyspark.sql.functions as F
from main.base import PySparkJobInterface

# Schema for both CSVs
plants_schema = StructType([
    StructField("plantId", StringType(), True),
    StructField("temperature", DoubleType(), True)
])


class PySparkJob(PySparkJobInterface):

    def init_spark_session(self) -> SparkSession:
        return (
            SparkSession.builder
            .master("local")
            .appName("Faulty Plants Detection")
            .getOrCreate()
        )

    def read_csv(self, input_path: str) -> DataFrame:
        return (
            self.spark.read
            .option("header", "false")
            .schema(plants_schema)
            .csv(input_path)
        )

    def calc_average_temperature(self, observed: DataFrame) -> DataFrame:
        return (
            observed.groupBy("plantId")
            .agg(F.avg("temperature").alias("temperature"))
        )

    def find_faulty_plants(self, avg_observed: DataFrame, required:
DataFrame) -> DataFrame:
        joined_df = avg_observed.join(required, "plantId")
```

```
        faulty_df = joined_df.withColumn(
            "diff",
            F.abs(F.col("avg_observed.temperature") -
F.col("required.temperature"))
        )

        return (
            joined_df.withColumn(
                "diff",
                F.abs(F.col("avg_observed.temperature") -
F.col("required.temperature"))
            )
            .filter(F.col("diff") >= 5)
            .select(
                F.col("plantId"),
                F.col("avg_observed.temperature").alias("temperature")
            )
        )

    def save_as(self, data: DataFrame, output_path: str) -> None:
        (
            data.select("plantId", "temperature")
            .write
            .mode("overwrite")
            .option("header", "false")
            .csv(output_path)
        )
```